

Question description

There is a bit string consisting of n bits. Then, there are some changes that invert one given bit. Your task is to report, after each change, the length of the longest substring whose each bit is the same.

Constraints

- $1 \leq n \leq 2 \cdot 10^5$
- $1 \leq m \leq 2 \cdot 10^5$
- $1 \leq x_i \leq n$

Input Format

The first input line has a bit string consisting of n bits. The bits are numbered $1, 2, \dots, n$.

The next line contains an integer m : the number of changes.

The last line contains m integers $x_1, x_2, \dots, x_m$ describing the changes.

Output Format

After each change, print the length of the longest substring whose each bit is the same.

▼ Logical Test Cases

Test Case 1

INPUT (STDIN)

001011
3
3 2 5

EXPECTED OUTPUT

4 2 3

Test Case 2

INPUT (STDIN)

101010
4
1 3 2 6

EXPECTED OUTPUT

2 4 2 2

▼ Mandatory Test Cases

Test Case 1

KEYWORD

void pull(int k,int
l,int r)

▼ Complexity Test Cases

Test Case 1

CYCLOMATIC COMPLEXITY

2

Test Case 2

TOKEN COUNT

550

Test Case 3

NLOC

59